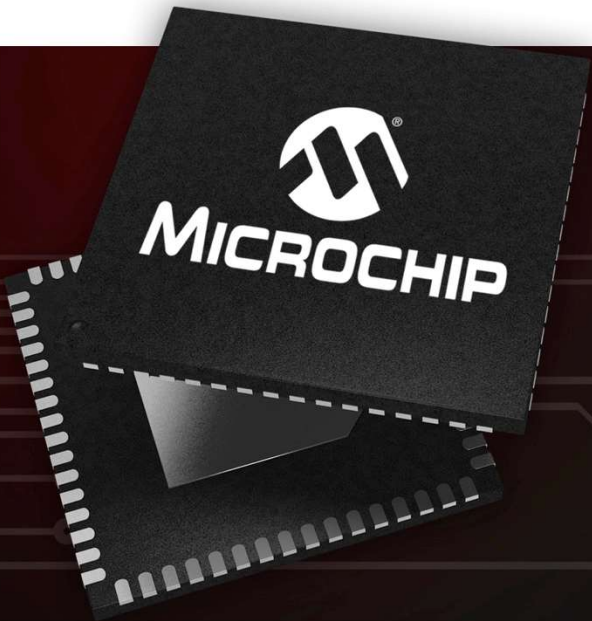




A Leading Provider of Microcontroller,
Mixed-Signal, Analog & Flash-IP Solutions



maXTouch Studio Lite使用说明

RZ



下载,安装maXTouch Studio Lite

进入microchip官方网址下载maxtouch-lite安装包后安装。

<https://www.microchip.com/en-us/products/touch-and-gesture/solutions-for-touchpads-and-touchscreens/maxtouch-touchscreen-controllers>

The screenshot shows the Microchip website's navigation bar with the Microchip logo and links for Products, Solutions, Tools and Software, Support, Education, About, and Order Now. Below the navigation bar, a breadcrumb trail is visible: Products / Touch and Gesture / Solutions for Touchpads and Touchscreens / maXTouch Touchscreen Controllers. The main content area features a dark blue background with a circuit pattern. On the left, there are links for ATMXT228UD-HA1 and ATMXT112UD-HA1. On the right, there is a video player showing a touch screen interface. Below the main content area, there are two white boxes. The left box is titled 'maXTouch® Studio' and contains text about the Integrated Development Platform (IDP) for developing and debugging with maXTouch touchscreen controllers. A red box highlights the 'Download the Latest Version of maXTouch Studio Lite' button. The right box is titled 'Would You Like to Use a Microcontroller to Create Your 2D Touch Surface Interface?' and contains text about integrating a touch interface into a microcontroller. A 'Learn More' button is highlighted with a red box.

maXTouch® Studio

Create, design and tune your next innovative touchscreen. maXTouch Studio LITE is the Integrated Development Platform (IDP) for developing and debugging with our maXTouch touchscreen controllers, and it is available for use without requiring a Non-Disclosure Agreement (NDA).

Download the Latest Version of maXTouch Studio Lite

Would You Like to Use a Microcontroller to Create Your 2D Touch Surface Interface?

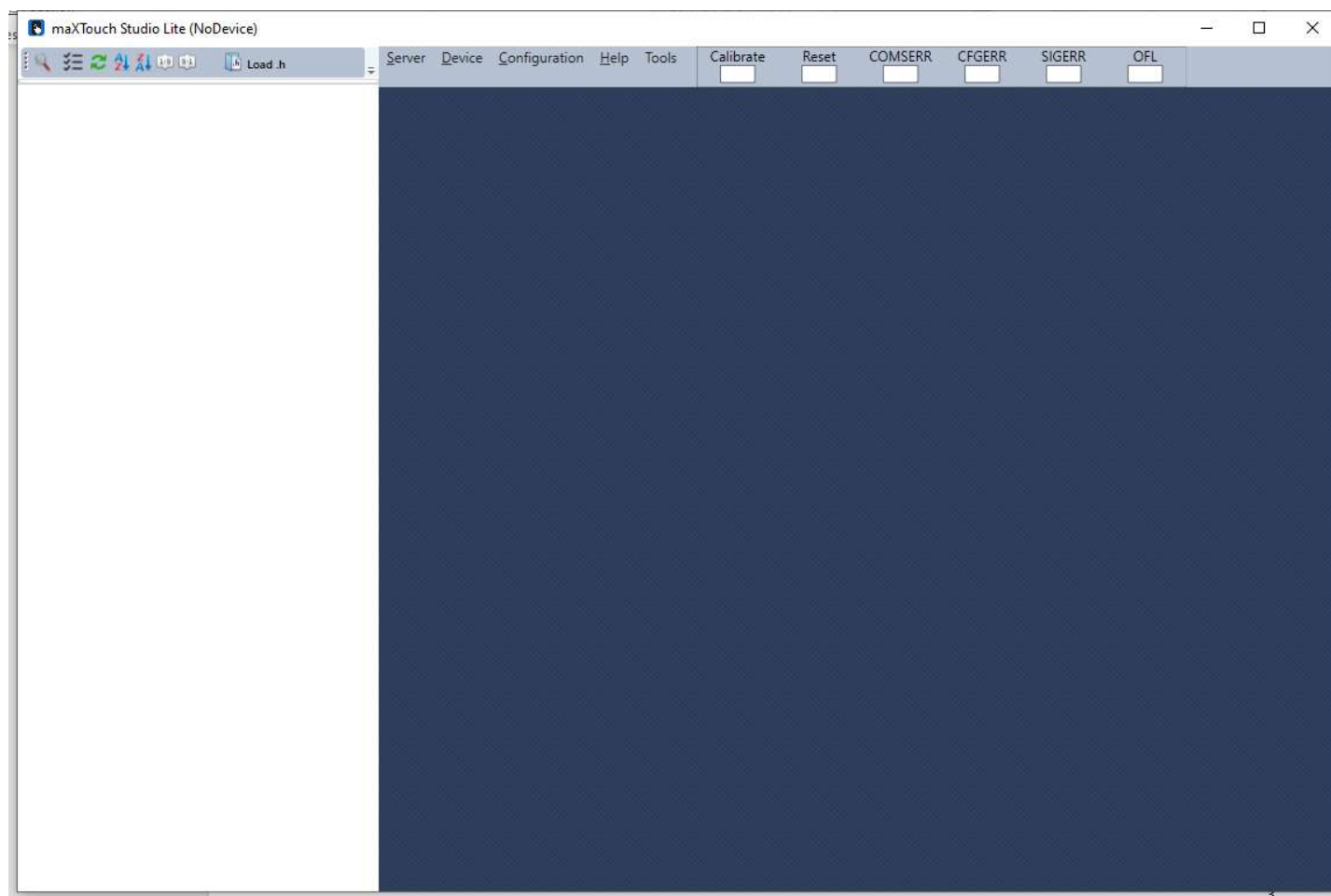
Discover how to cost-efficiently integrate a touch interface into the microcontroller of your choice using our 2D Touch Surface Library.

Learn More



maXTouch Studio Lite使用说明

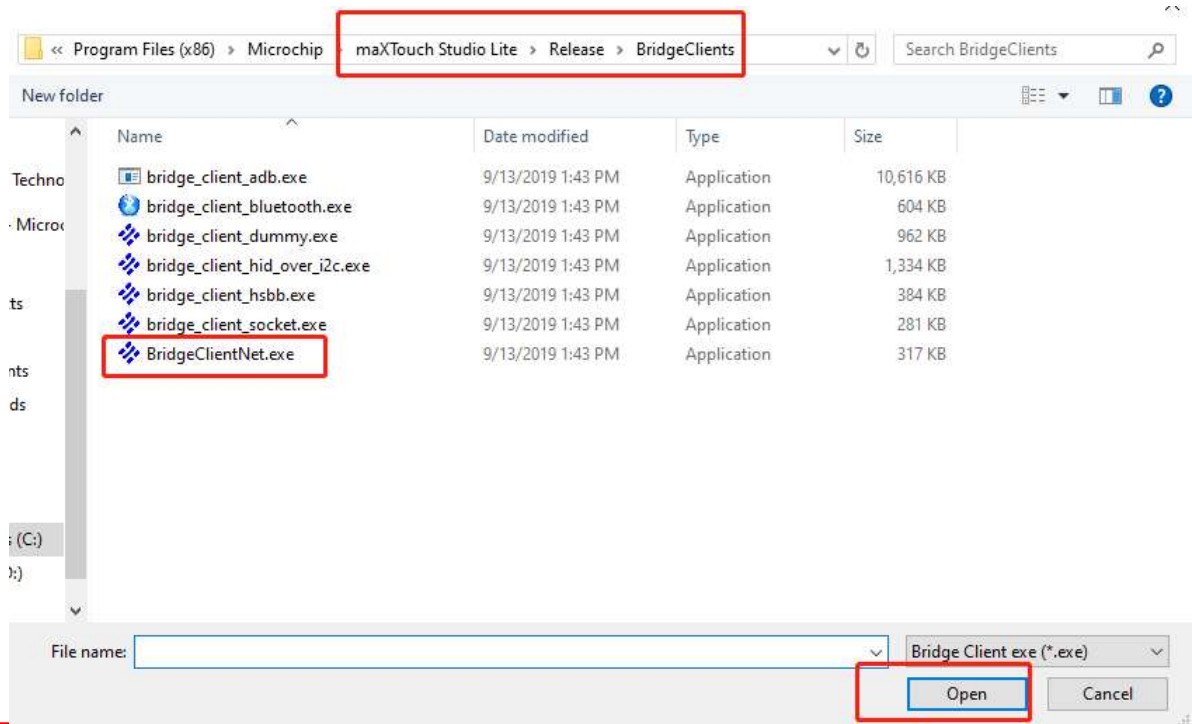
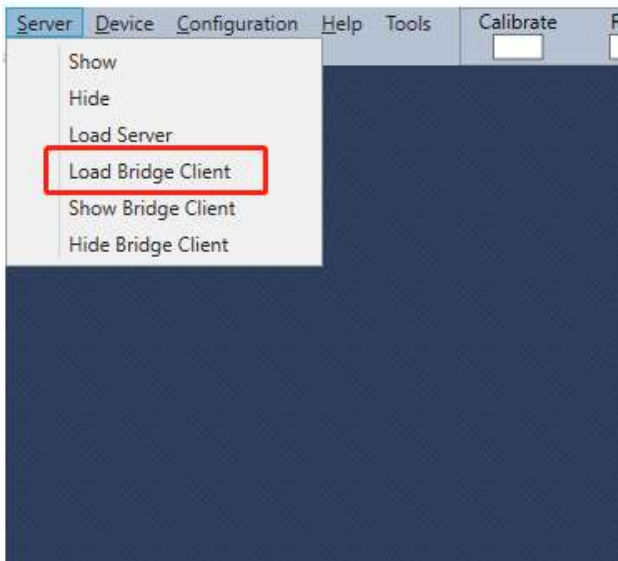
打开maXTouch Studio Lite后界面如下



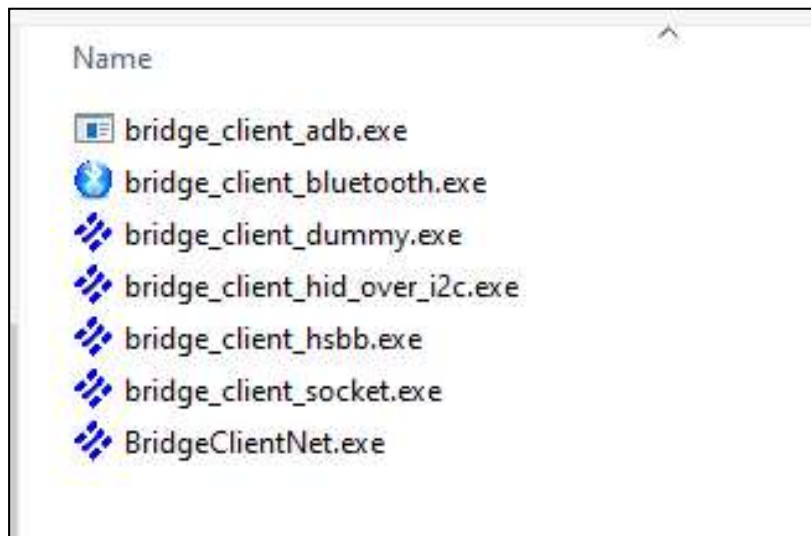
maXTouch Studio Lite使用说明

根据不同的连接方式,Load不同的Bridge client 进行数据连接.

以下以测试板为例,
将产品接上测试板后通过USB连接电脑,注意确认TP供电连接正常.
点击Server选择Load Bridge Client,选择安装目录/Release/bridgeClients/下对应的
bridge client. 测试板直接连TP的情况如下图选择即可.



maXTouch Studio Lite使用说明



Bridge client说明:

bridge_client_adb.exe:

通过ADB连接(android),主机需要root权限.

bridge_client_socket.exe:

通过网络连接(linux),主机需要root权限

bridge_client_hid_over_i2c.exe:

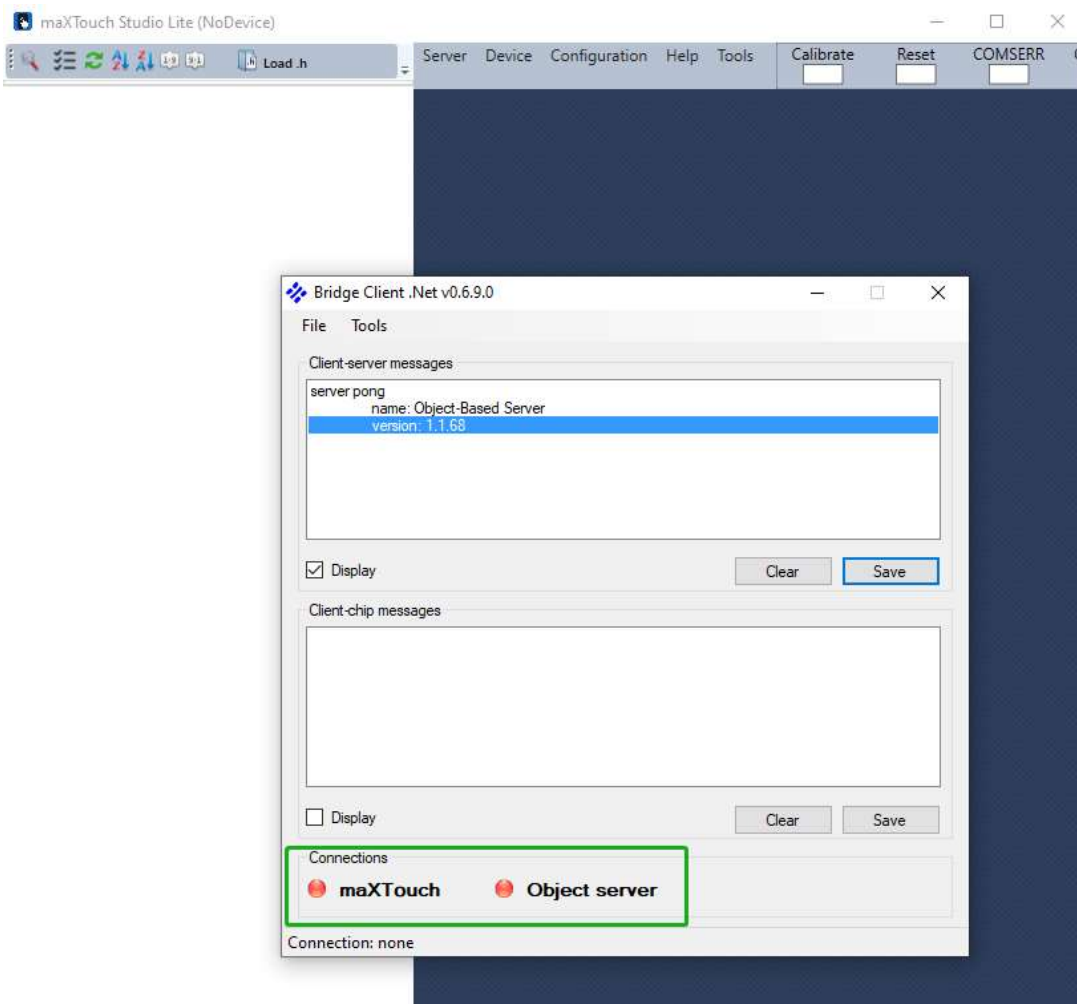
bridge_client_hsbb.exe:

BridgeClientNet.exe

(或bridge_client_5030.exe) :

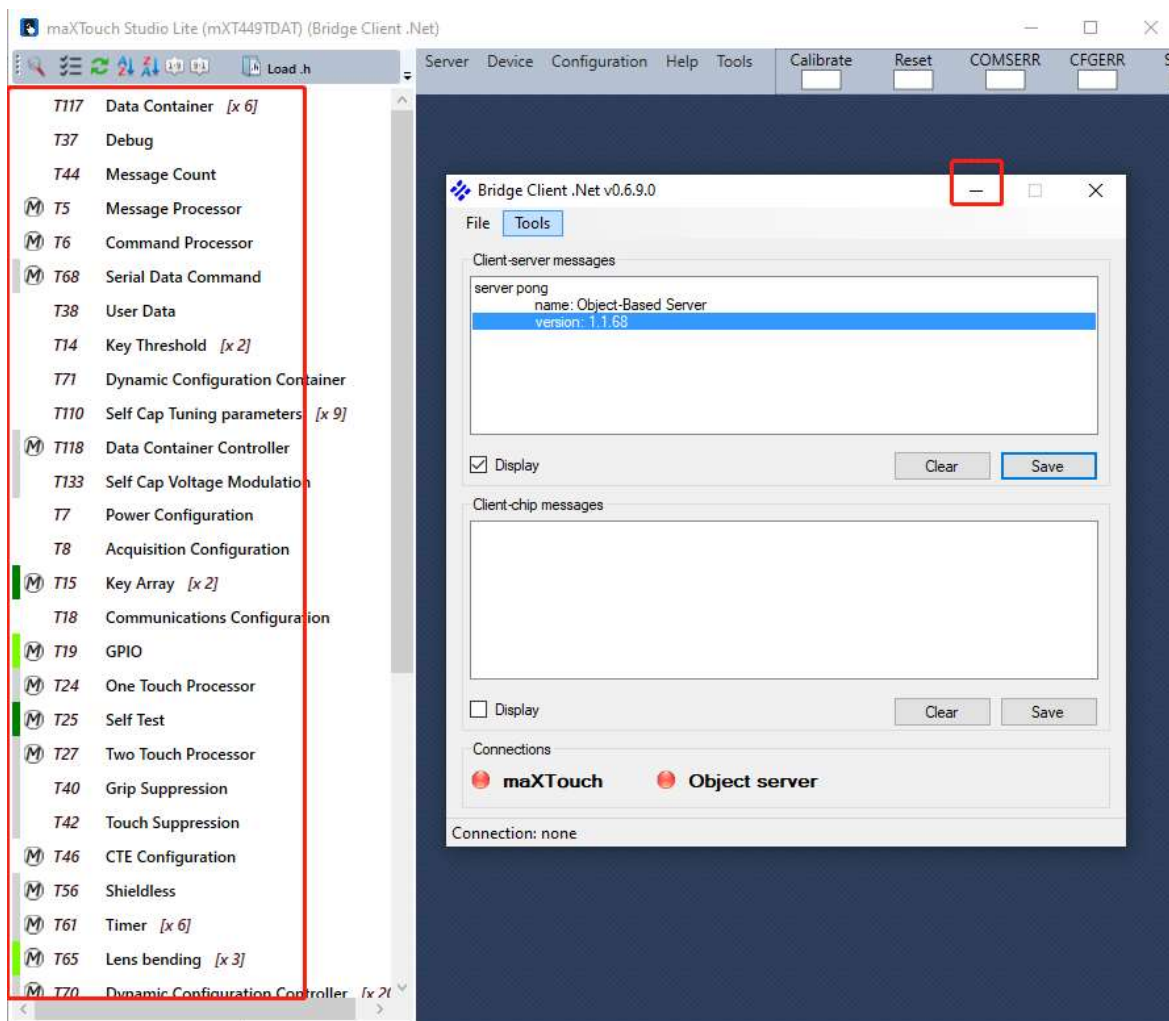
通过测试板连接.(直连TP)

打开Bridge Client后等待连接. 确保测试板,TP连接正常.



maXTouch Studio Lite使用说明

连接成功后,软件左侧会有TP参数列表显示,点击对应的参数,可进行参数调整.



注意: 连接成功后不要
将Bridge Client关闭,最
小化即可,否则连接会中
断!!!

maXTouch Studio Lite使用说明

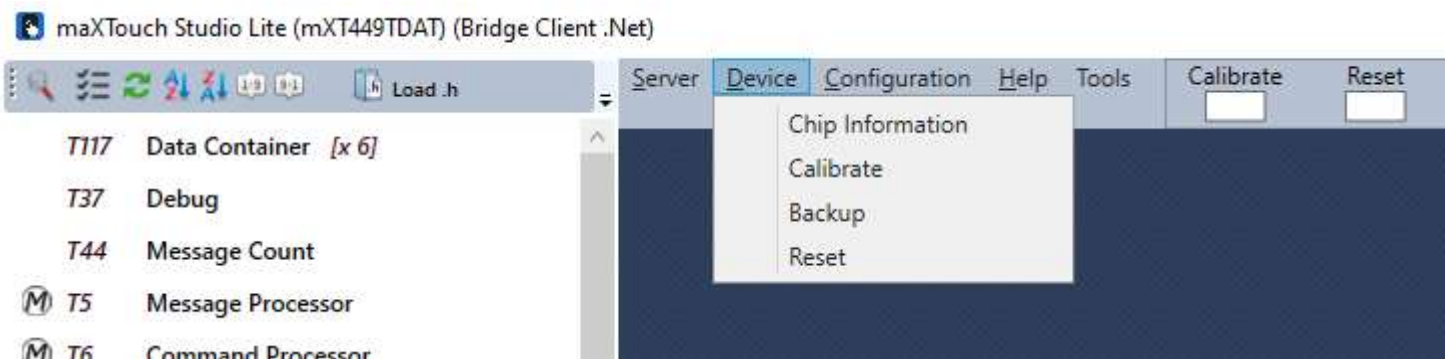
Device操作说明:

Chip Information: 查看芯片信息.

Calibrate: 信号校准

Backup: 将配置存储到芯片的NVM区, 掉电后修改的参数不会丢失.

Reset: 重启芯片



注意: 一般的参数修改会暂存在芯片的RAM区, 对应的功能会生效, 掉电/reset之后会恢复成原有参数. 执行backup之后参数会更新到芯片NVM区, 掉电/reset后会保持修改后的参数.

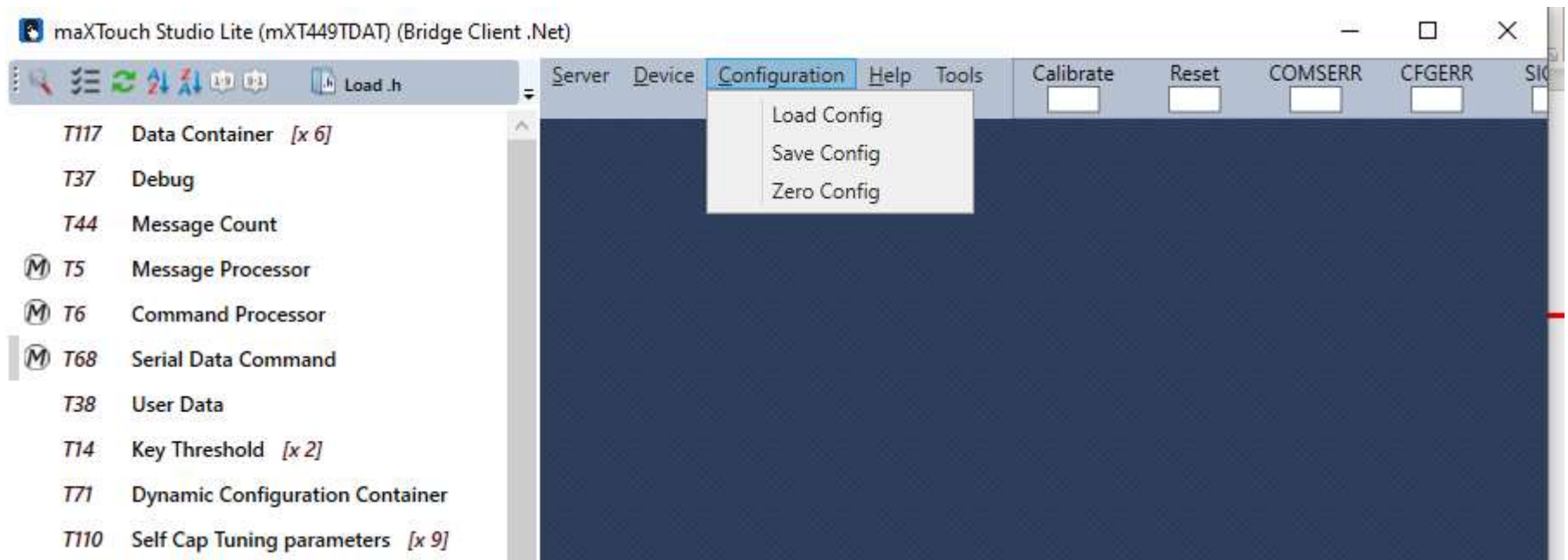
maXTouch Studio Lite使用说明

Configuration操作说明:

Load Config: 从文件导入配置文件(支持xcfg和raw文件).
如需保存到芯片中,需要执行backup操作.

Save Config: 将当前芯片中的配置存储成文件(支持xcfg和raw文件).
如有参数调整,需先执行backup后再save.

Zero Config: 所有配置清零.



Tools操作说明:

Basic Settings:

Graphical Debug Viewer:

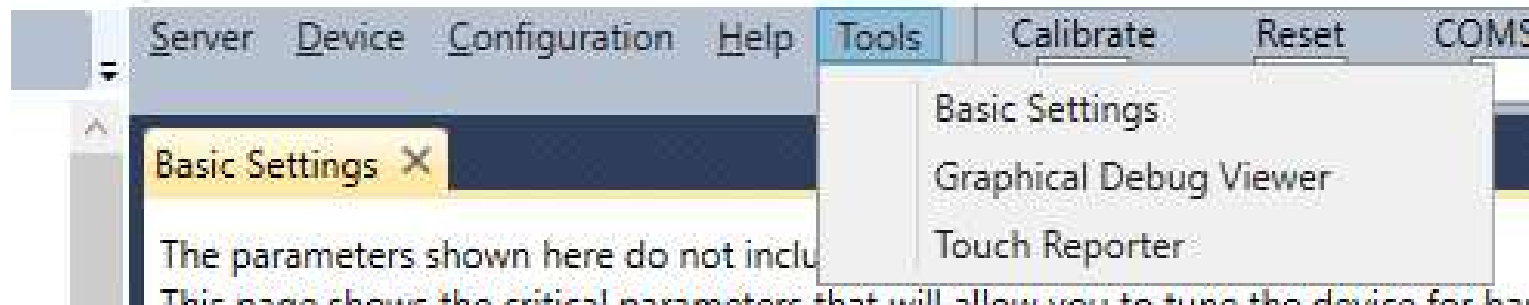
Touch Report:

基本参数设置.

TP数据图形化调试,参考下一页.

TP报点查看

lient .Net)

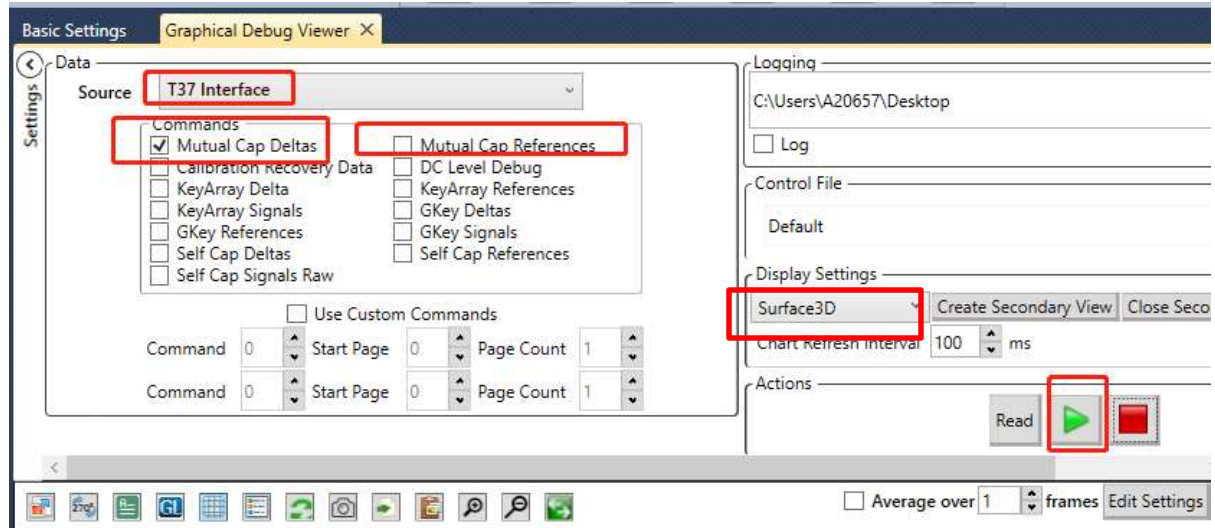


maXTouch Studio Lite使用说明

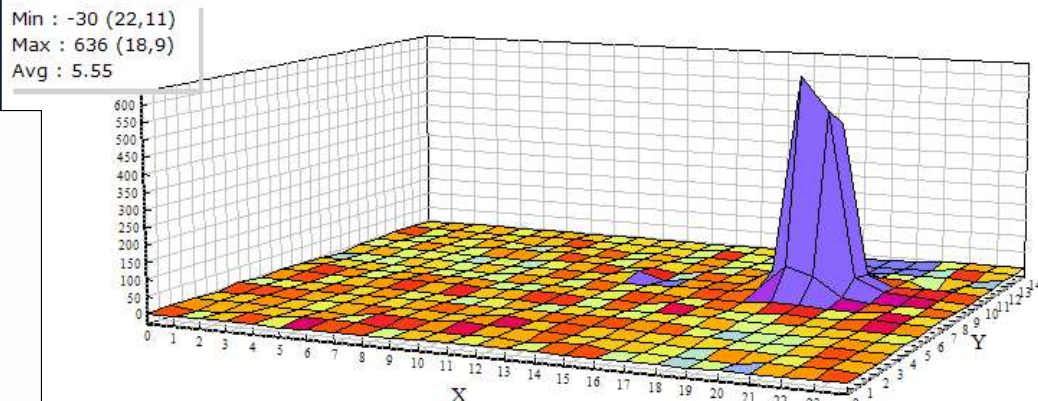
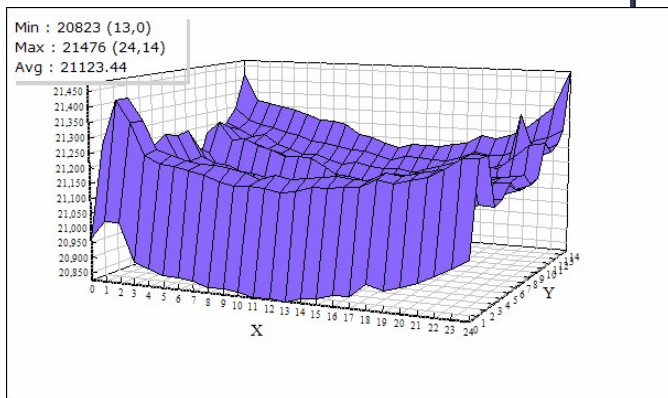
Graphical Debug Viewer:
选择不同的commands查看
对应的动态数据.

Mutual Cap Deltas:
互容触摸信号Delta.

Mutual Cap References:
互容触摸信号Reference.



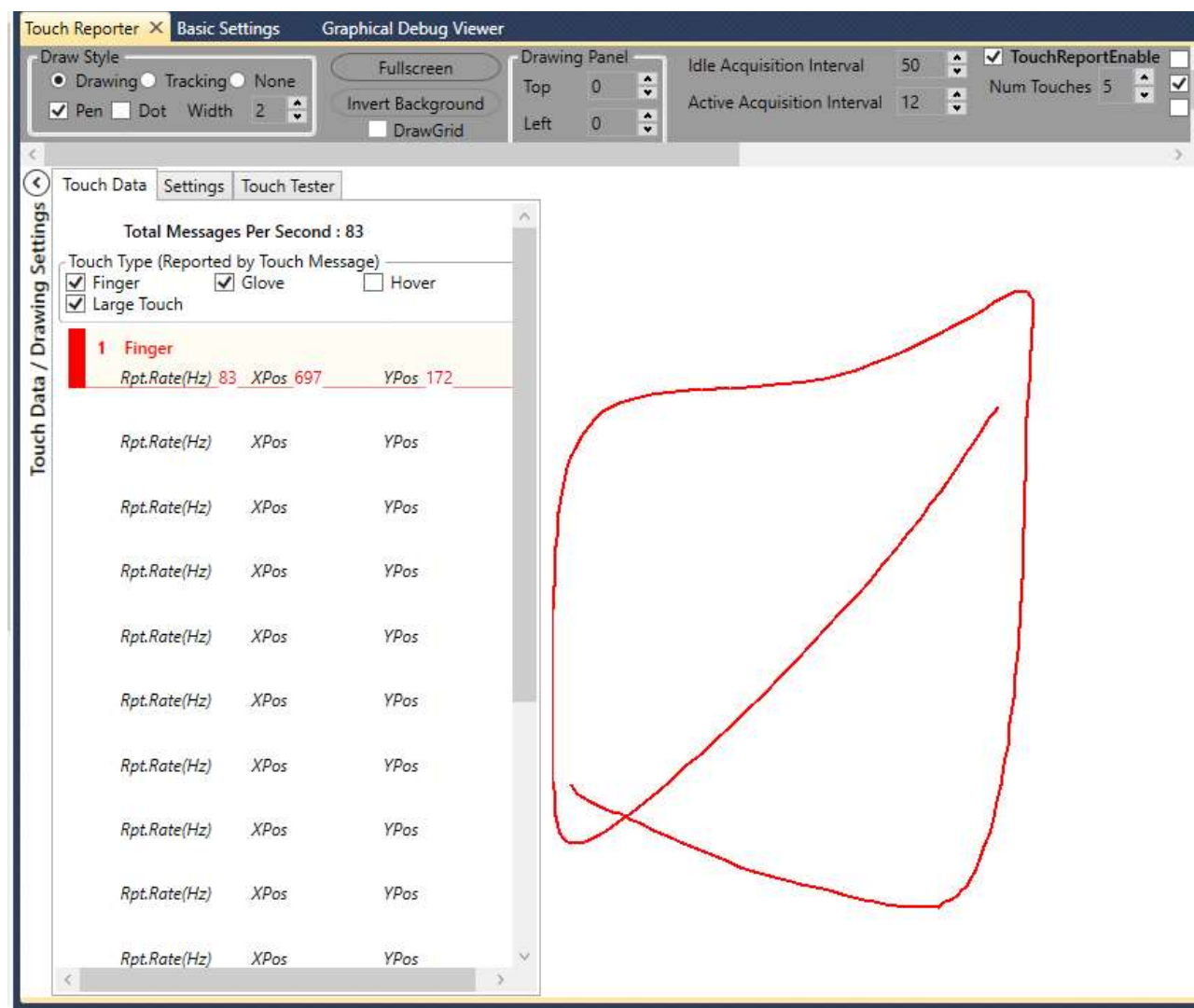
Mutual Cap References



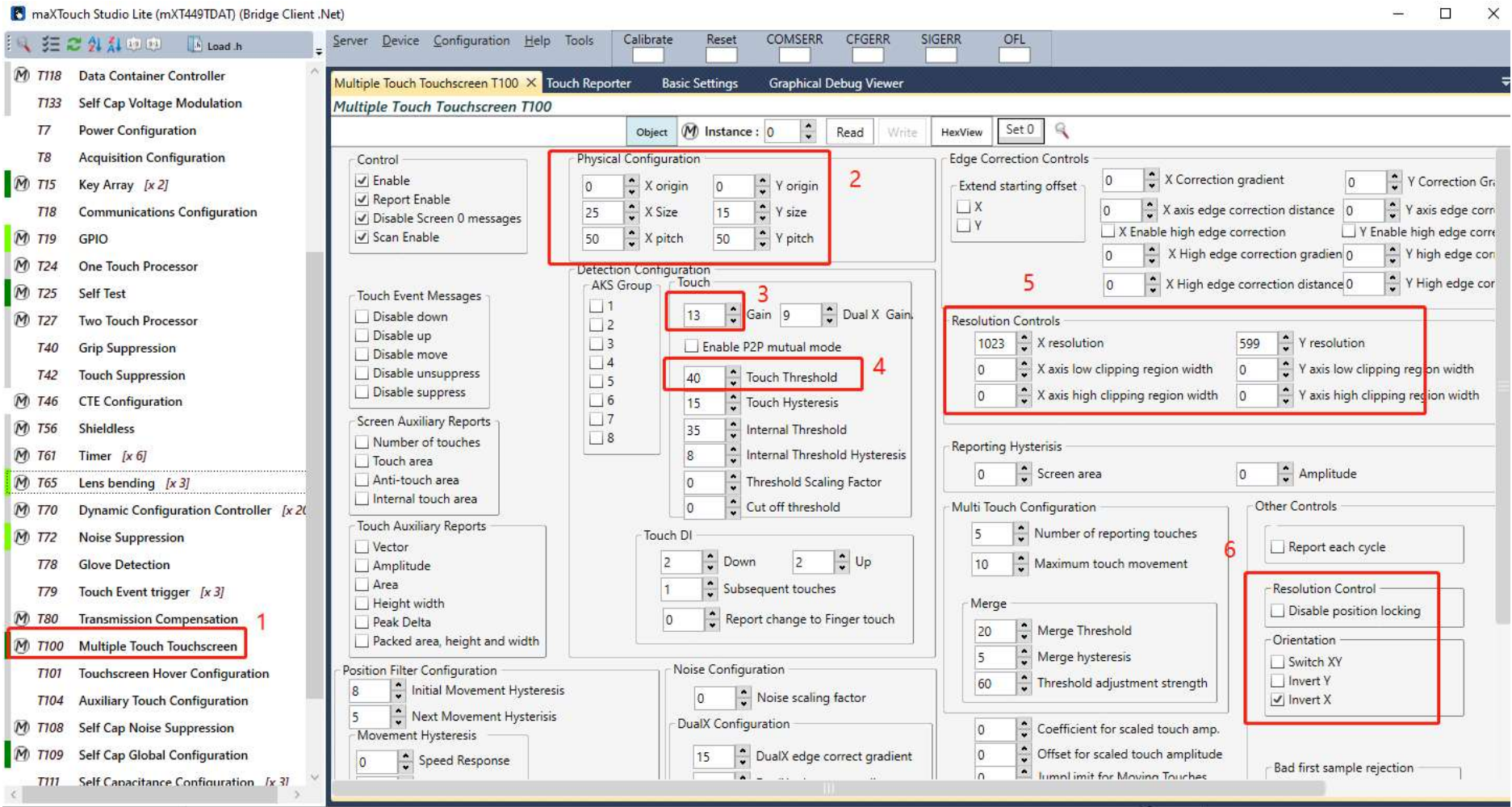
Mutual Cap Deltas

maXTouch Studio Lite使用说明

Touch Report:
查看TP报点是否正常.



T100参数修改示例:



The screenshot shows the maXTouch Studio Lite interface with the following configuration details:

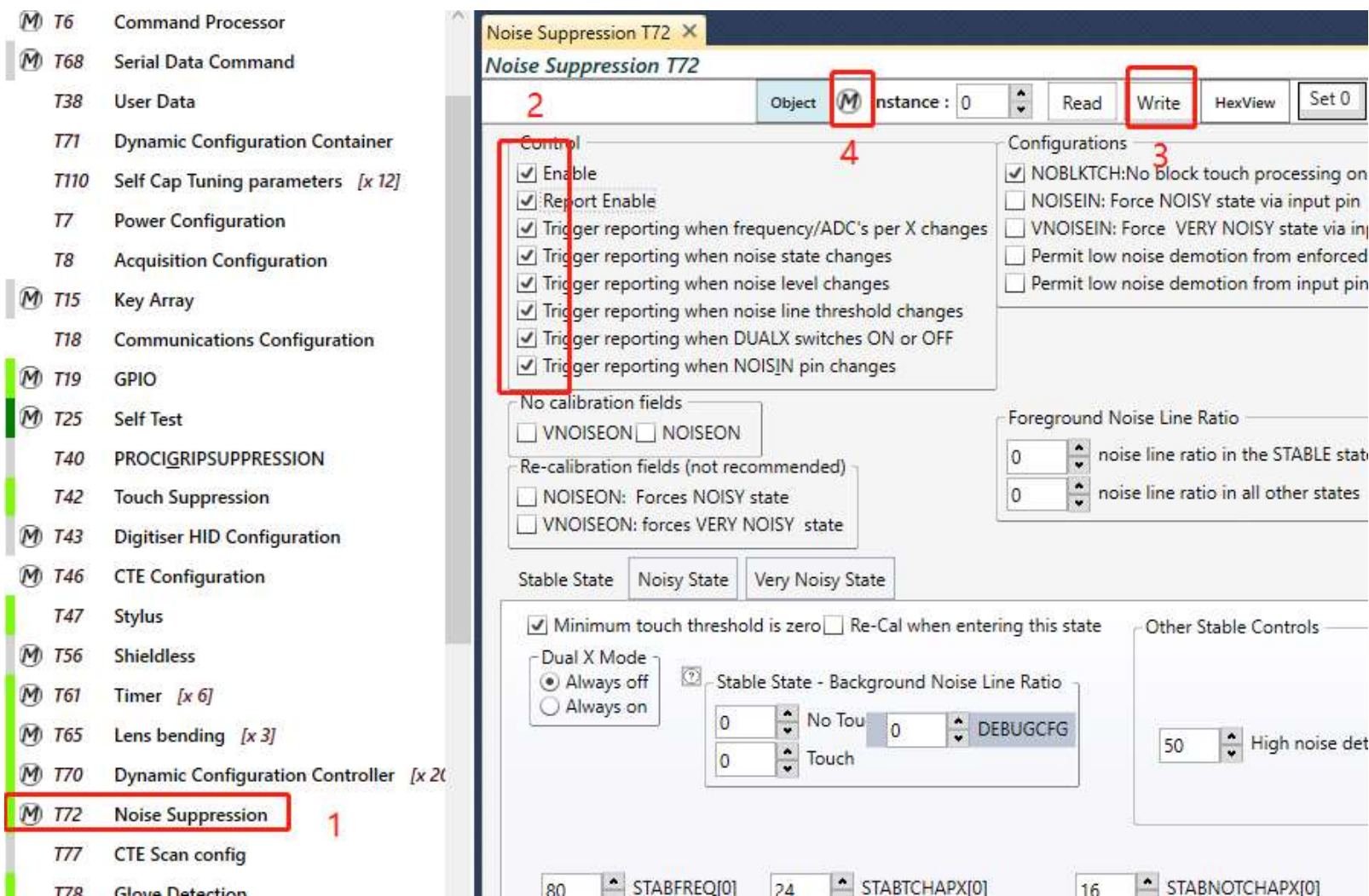
- Left Sidebar (Tree View):**
 - T118 Data Container Controller
 - T133 Self Cap Voltage Modulation
 - T7 Power Configuration
 - T8 Acquisition Configuration
 - T115 Key Array [x 2]**
 - T18 Communications Configuration
 - T19 GPIO
 - T24 One Touch Processor
 - T25 Self Test
 - T27 Two Touch Processor
 - T40 Grip Suppression
 - T42 Touch Suppression
 - T46 CTE Configuration
 - T56 Shieldless
 - T61 Timer [x 6]
 - T65 Lens bending [x 3]
 - T70 Dynamic Configuration Controller [x 2]
 - T72 Noise Suppression
 - T78 Glove Detection
 - T79 Touch Event trigger [x 3]
 - T80 Transmission Compensation 1**
 - T100 Multiple Touch Touchscreen**
 - T101 Touchscreen Hover Configuration
 - T104 Auxiliary Touch Configuration
 - T108 Self Cap Noise Suppression
 - T109 Self Cap Global Configuration
 - T111 Self Capacitance Configuration [x 3]
- Main Panel (Multiple Touch Touchscreen T100):**
 - Control:**
 - ☒ Enable
 - ☒ Report Enable
 - ☒ Disable Screen 0 messages
 - ☒ Scan Enable
 - Physical Configuration:**
 - X origin: 0
 - X Size: 25
 - X pitch: 50
 - Y origin: 0
 - Y size: 15
 - Y pitch: 50
 - Detection Configuration:**
 - AKS Group: 13
 - Gain: 9
 - ☐ Enable P2P mutual mode
 - Touch Threshold: 40
 - Touch Hysteresis: 15
 - Internal Threshold: 35
 - Internal Threshold Hysteresis: 8
 - Threshold Scaling Factor: 0
 - Cut off threshold: 0
 - Touch DI:**
 - Down: 2
 - Up: 2
 - Subsequent touches: 1
 - Report change to Finger touch: 0
 - Position Filter Configuration:**
 - Initial Movement Hysteresis: 8
 - Next Movement Hysteresis: 5
 - Movement Hysteresis: 0
 - Speed Response: 0
 - Noise Configuration:**
 - Noise scaling factor: 0
 - DualX Configuration: 15
 - Edge Correction Controls:**
 - Extend starting offset: ☐ X, ☐ Y
 - X Correction gradient: 0
 - X axis edge correction distance: 0
 - X High edge correction gradient: 0
 - X High edge correction distance: 0
 - Y Correction gradient: 0
 - Y axis edge correction distance: 0
 - Y High edge correction gradient: 0
 - Y High edge correction distance: 0
 - Resolution Controls:**
 - X resolution: 1023
 - Y resolution: 599
 - X axis low clipping region width: 0
 - Y axis low clipping region width: 0
 - X axis high clipping region width: 0
 - Y axis high clipping region width: 0
 - Reporting Hysteresis:**
 - Screen area: 0
 - Amplitude: 0
 - Multi Touch Configuration:**
 - Number of reporting touches: 5
 - Maximum touch movement: 10
 - Merge:
 - Merge Threshold: 20
 - Merge hysteresis: 5
 - Threshold adjustment strength: 60
 - Coefficient for scaled touch amp: 0
 - Offset for scaled touch amplitude: 0
 - Limit limit for Moving Touches: 0
 - Other Controls:**
 - ☐ Report each cycle
 - Resolution Control:
 - ☐ Disable position locking
 - Orientation:
 - ☐ Switch XY
 - ☐ Invert Y
 - ☒ Invert X
 - Bad first sample rejection: ☐

T100参数修改示例:

1. 双击左侧T100进入参数页面
2. 修改通道数, origin为起始通道(一般为0), Size为通道数.
3. 修改Gain. 将Mutual Cap References调整到25000左右, Mutual Cap Delta调整到1000左右.
4. 调整TP报点的触发门限, 参考手指触摸时Mutual Cap Deltas大小, 设置到40%左右. 比如Delta信号为1000, 那设置值为 $1000 \times 0.4 / 8 = 50$. (芯片使用这个参数数时会内部会乘以8)
5. 分辨率调整, 根据Display要求来设置. 注意长短边与TP X/Y通道数的大小对应.
6. X/Y方向反转, 根据实际要求, 将坐标原点与方向调整到与产品要求一致, 可通过touch report来检视.

maXTouch Studio Lite使用说明

Log T72 noise 示例:



Left Sidebar (Tree View):

- M T6 Command Processor
- M T68 Serial Data Command
- T38 User Data
- T71 Dynamic Configuration Container
- T110 Self Cap Tuning parameters [x 12]
- T7 Power Configuration
- T8 Acquisition Configuration
- M T15 Key Array
- T18 Communications Configuration
- M T19 GPIO
- M T25 Self Test
- T40 PROCIGRIPSUPPRESSION
- T42 Touch Suppression
- M T43 Digitiser HID Configuration
- M T46 CTE Configuration
- T47 Stylus
- M T56 Shieldless
- M T61 Timer [x 6]
- M T65 Lens bending [x 3]
- M T70 Dynamic Configuration Controller [x 20]
- M T72 Noise Suppression** 1
- T77 CTE Scan config
- T78 Glove Detection

Noise Suppression T72 Configuration Window:

Object **M** Instance: 0 Read **Write** HexView Set 0

Control 2

- ☒ Enable
- ☒ Report Enable
- ☒ Trigger reporting when frequency/ADC's per X changes
- ☒ Trigger reporting when noise state changes
- ☒ Trigger reporting when noise level changes
- ☒ Trigger reporting when noise line threshold changes
- ☒ Trigger reporting when DUALX switches ON or OFF
- ☒ Trigger reporting when NOISEIN pin changes

Configurations 3

- ☒ NOBLKTCH: No block touch processing on
- ☐ NOISEIN: Force NOISY state via input pin
- ☐ VNOISEIN: Force VERY NOISY state via input pin
- ☐ Permit low noise demotion from enforced
- ☐ Permit low noise demotion from input pin

Foreground Noise Line Ratio

0 noise line ratio in the STABLE state

0 noise line ratio in all other states

Stable State **Noisy State** **Very Noisy State**

☒ Minimum touch threshold is zero ☐ Re-Cal when entering this state

Dual X Mode

- ☒ Always off
- ☐ Always on

Stable State - Background Noise Line Ratio

0 No Touch 0 DEBUGCFG

0 Touch

Other Stable Controls

50 High noise detection

80 STABFREQ[0] 24 STABTCHAPX[0] 16 STABNOTCHAPX[0]

maXTouch Studio Lite使用说明

Log T72 noise 示例:

1. 双击左侧T72进入参数页面
2. 确保勾选所有项.
3. 点击Write 写入修改.
4. 点击“M” 进入消息显示界面,如下图, log data可以抓取实时消息。

T72Messages

Logging
☒ Log Data
☐ Include Raw Data

Display
☐ Chart
☒ Text
 Clear Msg

Decoder
 Load custom decoder Load default decoder Save current decoder

☐ Filter Messages :
☐ Hexadecimal Display Number of rows **5**

消息显示区

ID	NOISEFINCHG	DUALXCHG	NITHRCHG	NOISEVLCHG	STATECHG	ACOCCHG	TCH	RLKCH	STATE	DUALX	NOISEFIN	ACOINDEX	MINTCHTHR	PKNOIS
38	0	0	0	1	0	0	0	0	STABLE	0	0	0	0	1
38	0	0	0	1	0	0	0	0	STABLE	0	0	0	0	1
38	0	0	0	1	0	0	0	0	STABLE	0	0	0	0	1
38	0	0	0	1	0	0	0	0	STABLE	0	0	0	0	0
38	0	0	1	0	1	0	0	0	STABLE	0	0	0	0	3

显示消息条数设置



Thank You!